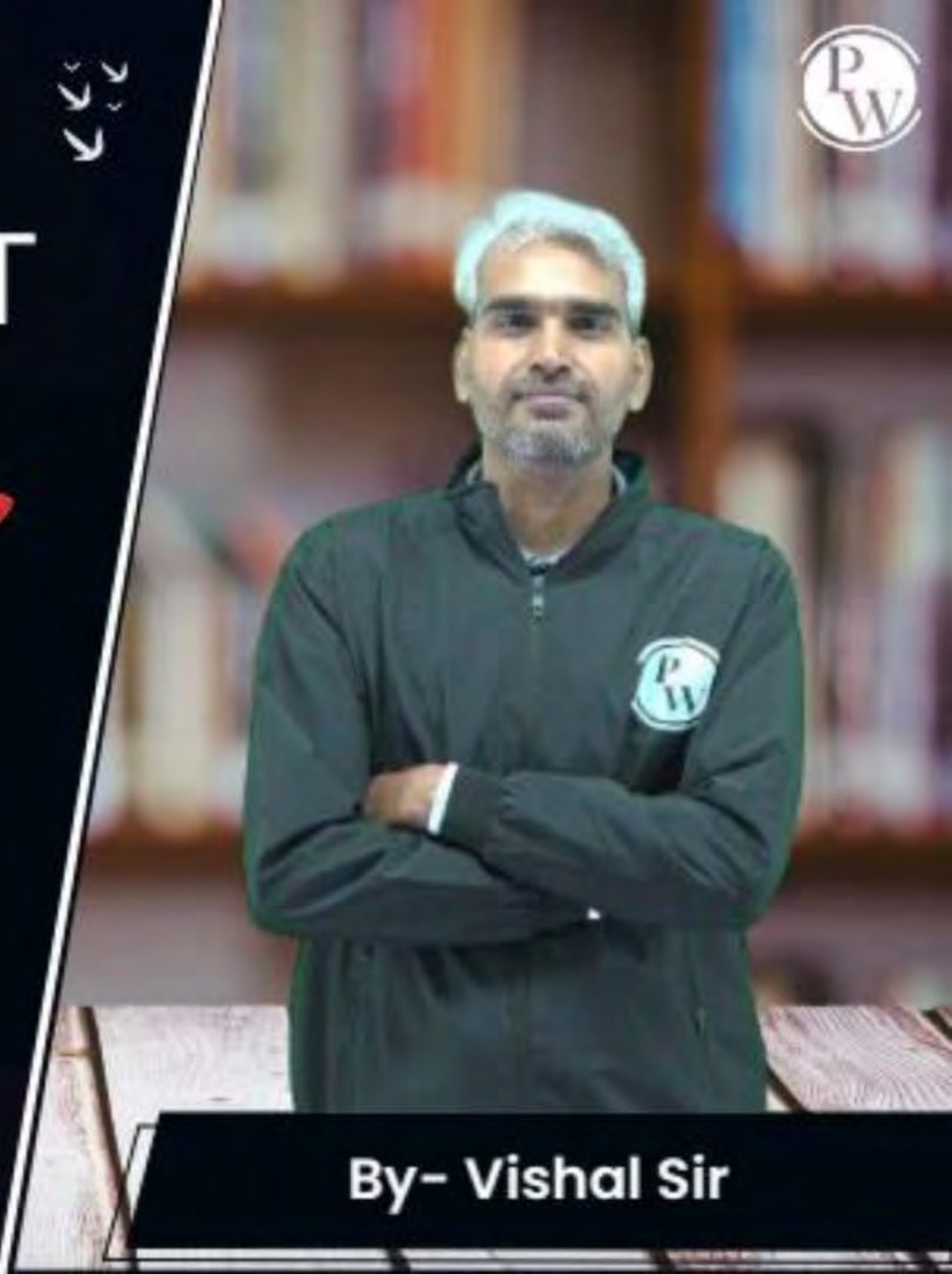
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 05



By- Vishal Sir

Recap of Previous Lecture



- Topic: Venn diagram
- Topic: Set operations

Slide

Topics to be Covered



Topic — Properties of set operations

Topic — Multi-set

Topic — Cartesian product

Slide



Topic : Venn Diagram

Venn diagram is used to represent the relationship among the sets pictorially.

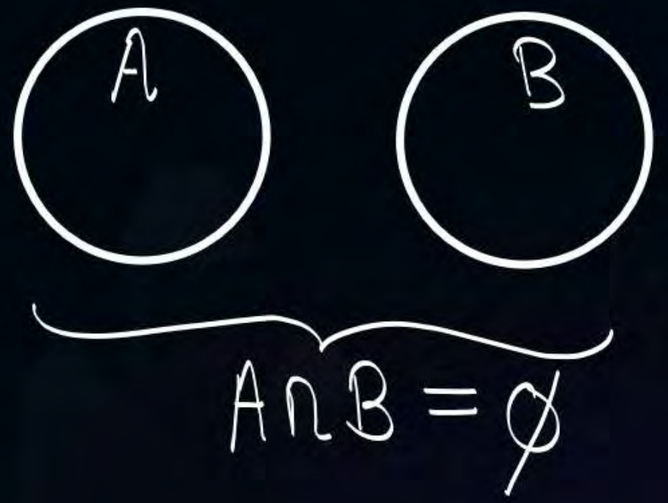


Here $B \subseteq A$.

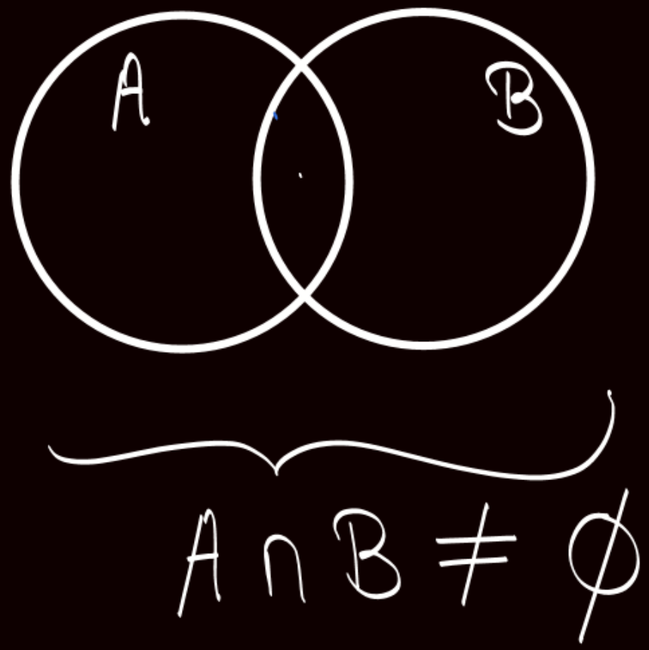
&

$B \cap A = B$

$B \cup A = A$



Slide



$A \cap B \neq \phi$



Topic : Set Operations

- Complement of a set $A^c = \{x \mid x \in U \text{ and } x \notin A\}$ | $A^c = U - A$
- Union of two sets $A \cup B = \{x \mid x \in A \text{ OR } x \in B\}$
- Intersection of two sets $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
- Set difference
  - $A - B = \{x \mid x \in A \text{ and } x \notin B\}$
  - $B - A = \{x \mid x \in B \text{ and } x \notin A\}$

$A \cap B = A - (A - B)$

$B \cap A = B - (B - A)$

- Symmetric difference of two sets $A \Delta B = \{x \mid (x \in A \text{ or } x \in B) \text{ and } x \notin (A \cap B)\}$

$A \Delta B = (A \cup B) - (A \cap B)$

$A \Delta B = (A - B) \cup (B - A)$

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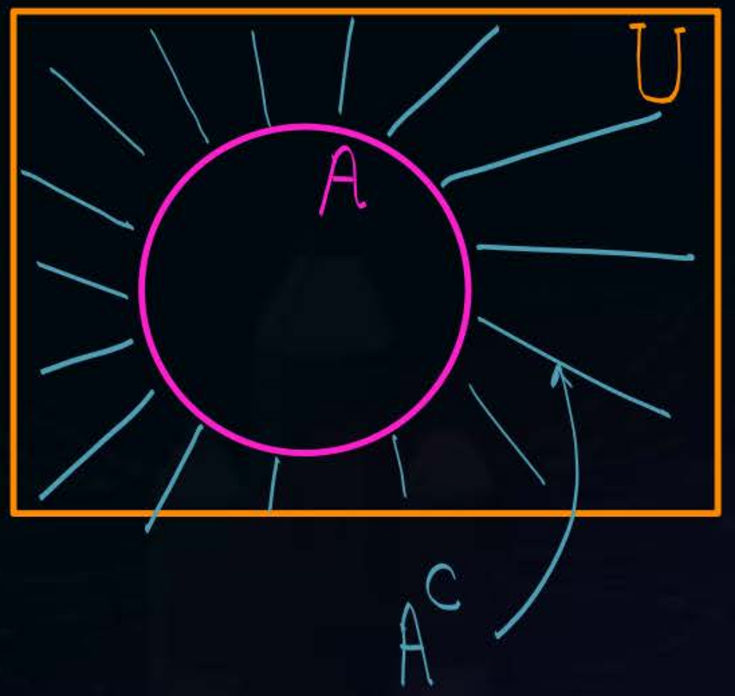


Topic : Complement of a set

Complement of a set is always defined w.r.t. universal set.

$$A^c = \{x \mid x \in U \text{ and } x \notin A\}$$

A^c → Complement of set A



$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$A = \{1, 2, 4, 7, 8, 9\}$

$A^c = \{3, 5, 6, 10\}$

Slide

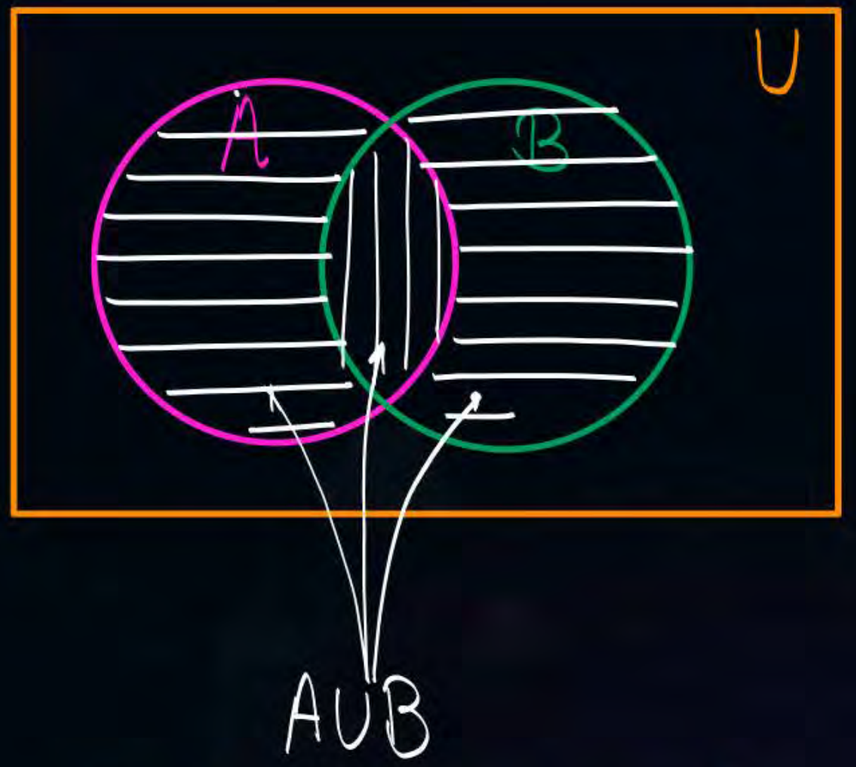


Topic : Union of two sets

Let A and B are two sets

then

$$A \cup B = \{x \mid x \in A \text{ OR } x \in B\}$$



Note: $A \cup B = B \cup A$

Slide

Topic : Intersection of two sets

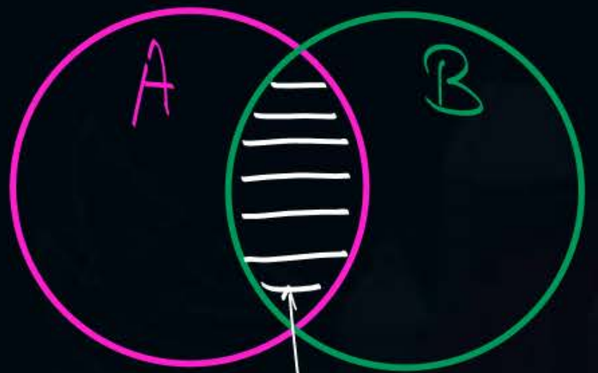
Let A & B are two sets

$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$



$A \cap B = \emptyset$

{If $A \cap B = \emptyset$ then A & B are called disjoint sets}



$A \cap B$

Note: $A \cap B = B \cap A$

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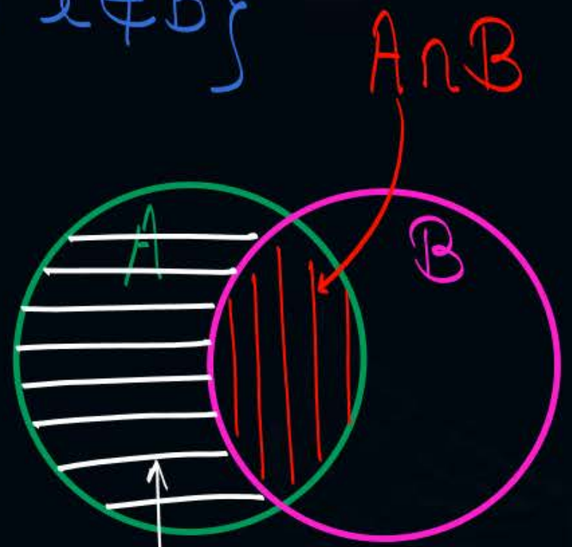


Topic : Set difference

let A & B are two sets

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

It will result all the elements that are present in set A, but not present in set B.



$$A - B = A - (A \cap B)$$

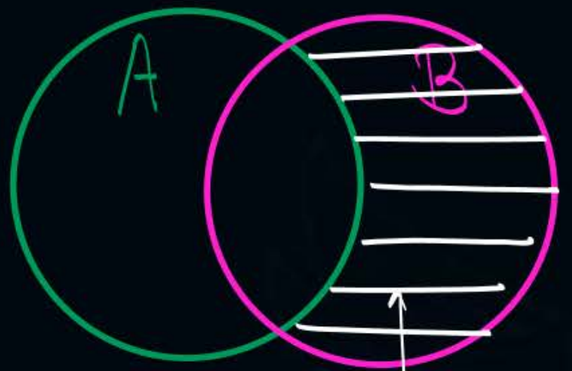
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Topic : Set difference

let A & B are two sets

$$B-A = \{x \mid x \in B \text{ and } x \notin A\}$$



$$B-A = B-(A \cap B)$$
$$= B-(B \cap A)$$

Slide

In general,

$$A - B \neq B - A$$

$A - B = B - A$ if and only if $A = B$,

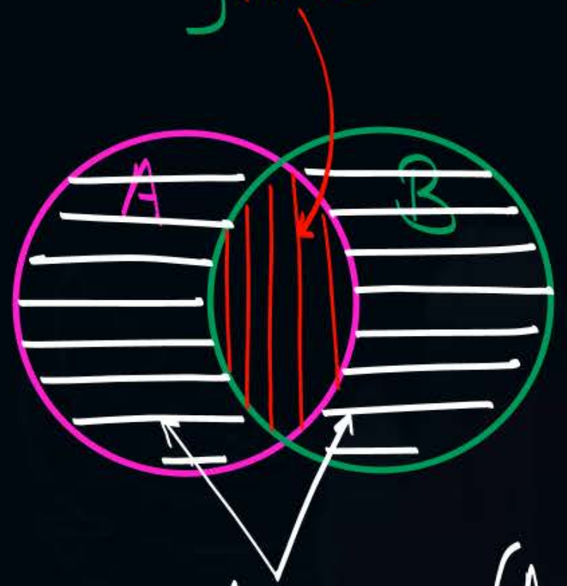
and in that case $A - B = B - A = \emptyset$



Topic : Symmetric difference of two sets

Let A & B are two sets

$$A \triangle B = \{x \mid (x \in A \text{ or } x \in B) \text{ and } x \notin A \cap B\}$$



$$A \triangle B = (A \cup B) - (A \cap B)$$

$$A \triangle B = (A - B) \cup (B - A)$$

Note: $A \triangle B = B \triangle A$

Slide



Topic : Properties of Set Operations

1. Idempotent:

   a. $A \cap A = A$

   b. $A \cup A = A$

2. Identity:

   a. $A \cup \emptyset = A$ { $\emptyset$ is identity w.r.t. Union }

   b. $A \cap U = A$ { U is identity w.r.t. intersection }

   ('Universal Set')

Slide



Topic : Properties of Set Operations

3. Domination:

a. $A \cap \emptyset = \emptyset$ { '$\emptyset$' dominates w.r.t. intersection }

b. $A \cup U = U$ { U dominates w.r.t. Union }

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Topic : Properties of Set Operations

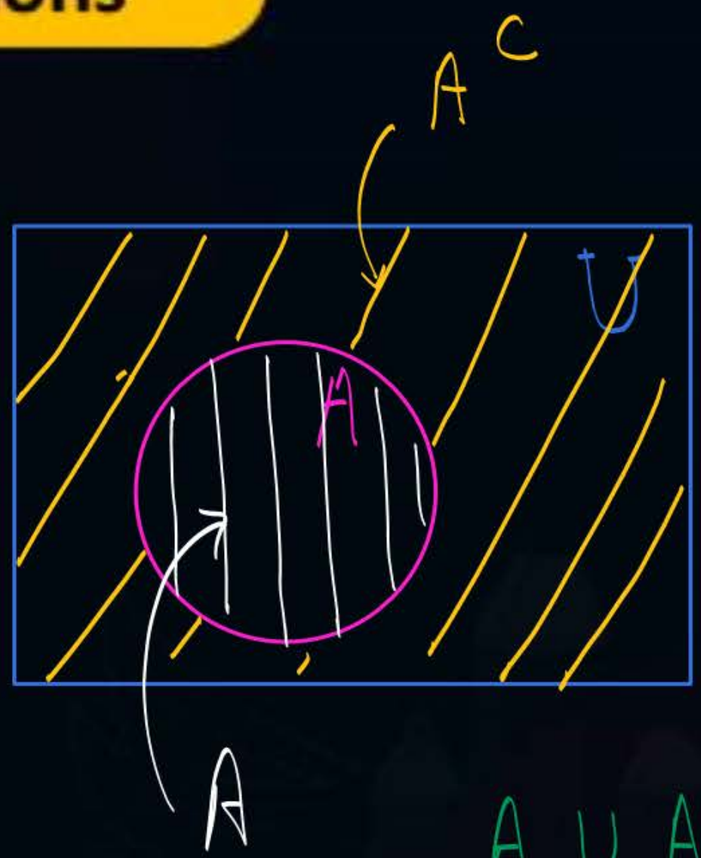
4. Complementation:

   a. $A \cup A^c = U$

   b. $A \cap A^c = \emptyset$

5. Double Complement:

   a. $(A^c)^c = A$



$A \cup A^c = U$

$A \cap A^c = \emptyset$



Topic : Properties of Set Operations

6. Commutative

   a. $A \cup B = B \cup A$

   b. $A \cap B = B \cap A$

7. Associative

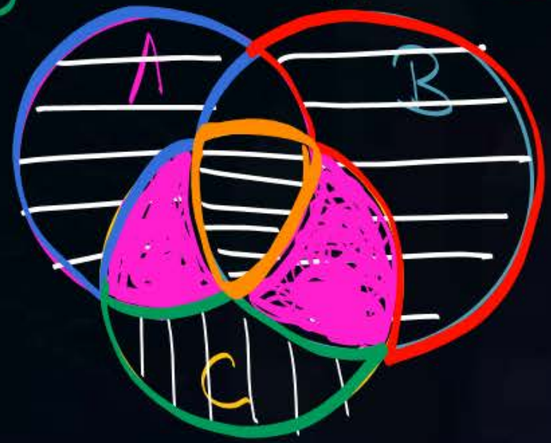
   a. $A \cup (B \cup C) = (A \cup B) \cup C$

   b. $A \cap (B \cap C) = (A \cap B) \cap C$

Slide

- Symmetric difference is Commutative
- $(A \triangle B) \triangle C = A \triangle (B \triangle C)$

∴ Symmetric difference is associative



$A \triangle (B \triangle C)$



$A = \{1, 2, 3\}$

$B = \{3\}$

$C = \{3\}$

$(A \Delta B) \Delta C$

$\{1, 2\} \Delta \{3\}$

$= \{1, 2, 3\}$

$A \Delta (B \Delta C)$

$\{1, 2, 3\} \Delta (\{3\} \Delta \{3\})$

$\{1, 2, 3\} \Delta \{\}$

$= \{1, 2, 3\}$

$A = \{1, 2, 3\}$

$B = \{3\}$

$C = \{2\}$

$(A \Delta B) \Delta C$

$\{1, 2\} \Delta \{2\}$

$= \{1\}$

$A \Delta (B \Delta C)$

$\{1, 2, 3\} \Delta (\{3\} \Delta \{2\})$

$\{1, 2, 3\} \Delta \{2, 3\}$

$= \{1\}$



Topic : Properties of Set Operations

8. Absorption

$A \cap B \subseteq A$

a. $A \cup (A \cap B) = A$ ✓

b. $A \cap (A \cup B) = A$ ✓

$A \cup B \supseteq A$

9. DeMorgan's

a. $(A \cup B)^c = A^c \cap B^c$

b. $(A \cap B)^c = A^c \cup B^c$

$(A^c \cap B^c)^c = A \cup B$

Slide



Topic : Properties of Set Operations

$\cap$ = $\cdot$

$\cup$ = $+$

10. Distributive

a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $\quad a + (b \cdot c) = (a+b)(a+c)$

b. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $\quad a \cdot (b+c) = (a \cdot b) + (a \cdot c)$

Slide

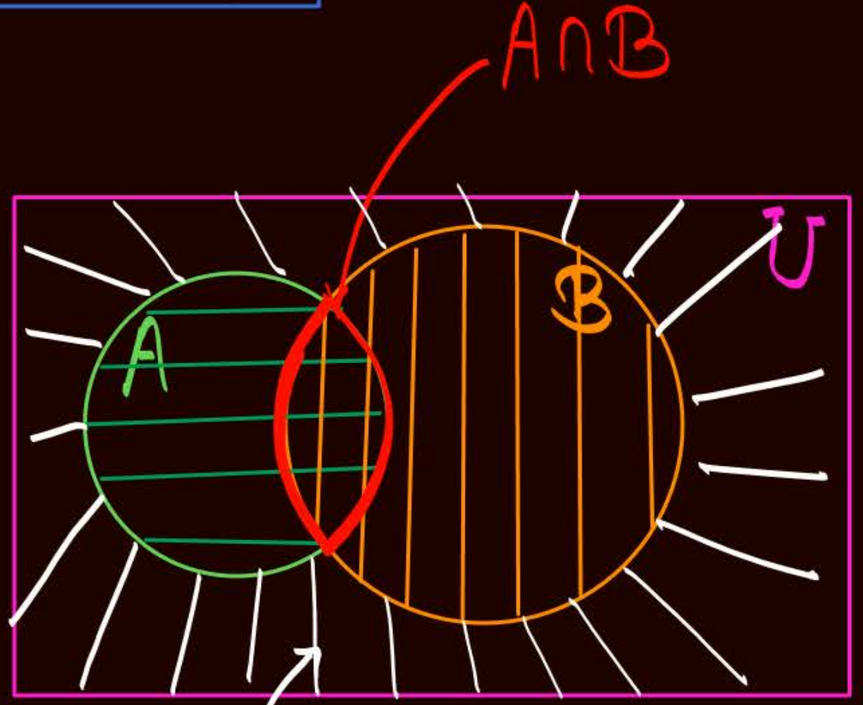
Principle of Inclusion Exclusion

$n(A)$ = No. of elements in set $A = |A|$

$n(B)$ = No. of elements in set $B = |B|$

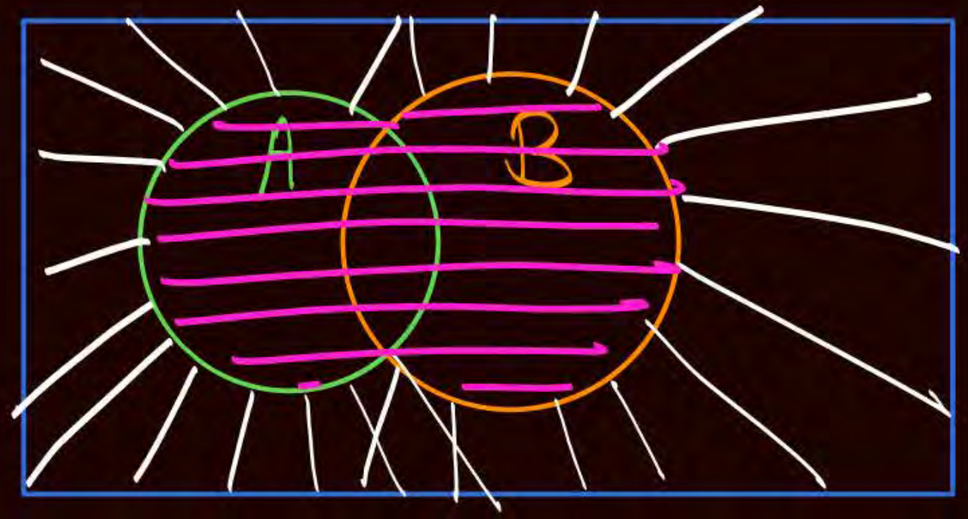
$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$



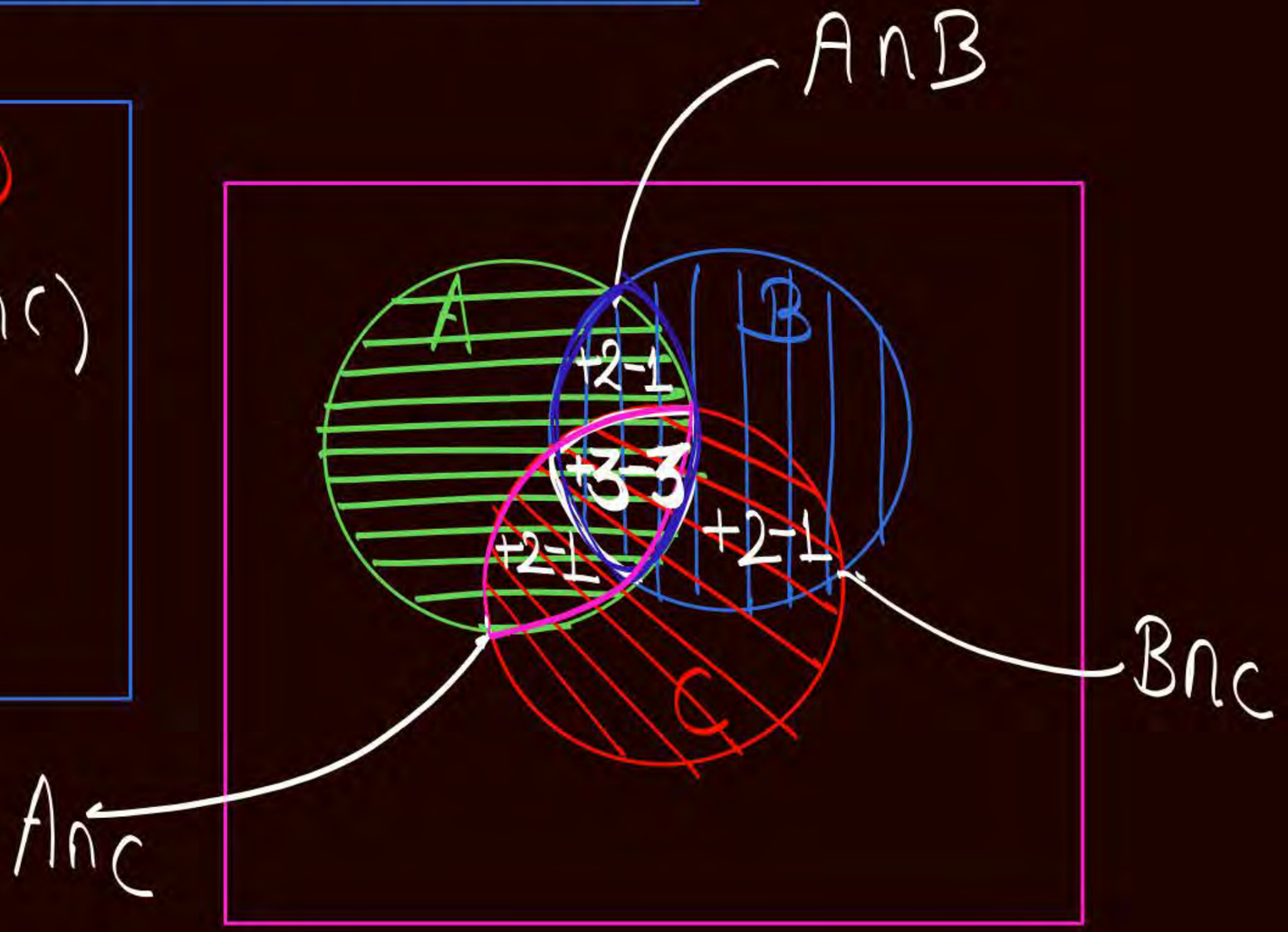
Elements that are neither Present in set A nor Present in set B

$$A^c \cap B^c = (A \cup B)^c = U - (A \cup B)$$


A
B

Principle of Inclusion Exclusion

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$





Topic : Multi-set

- An unordered Collection of objects/elements in which an element may appear more than once is Called a multiset.

eg. Multiset = $\{a, b, b, a, a, c, d, d\}$

(or frequency) = $\{3.a, 2.b, 1.c, 2.d\}$

These numbers represent the multiplicity of Corresponding element

Slide

Let multiset $P = \{ m_1.a_1, m_2.a_2, m_3 a_3, \ldots\ldots, m_k.a_k \}$

where 'm_i' is the multiplicity of element a_i.

And

let multiset $Q = \{ n_1.a_1, n_2.a_2, n_3.a_3, \ldots\ldots, n_k.a_k \}$

where 'n_i' is the multiplicity of element a_i.

Find multiplicity of a_i in

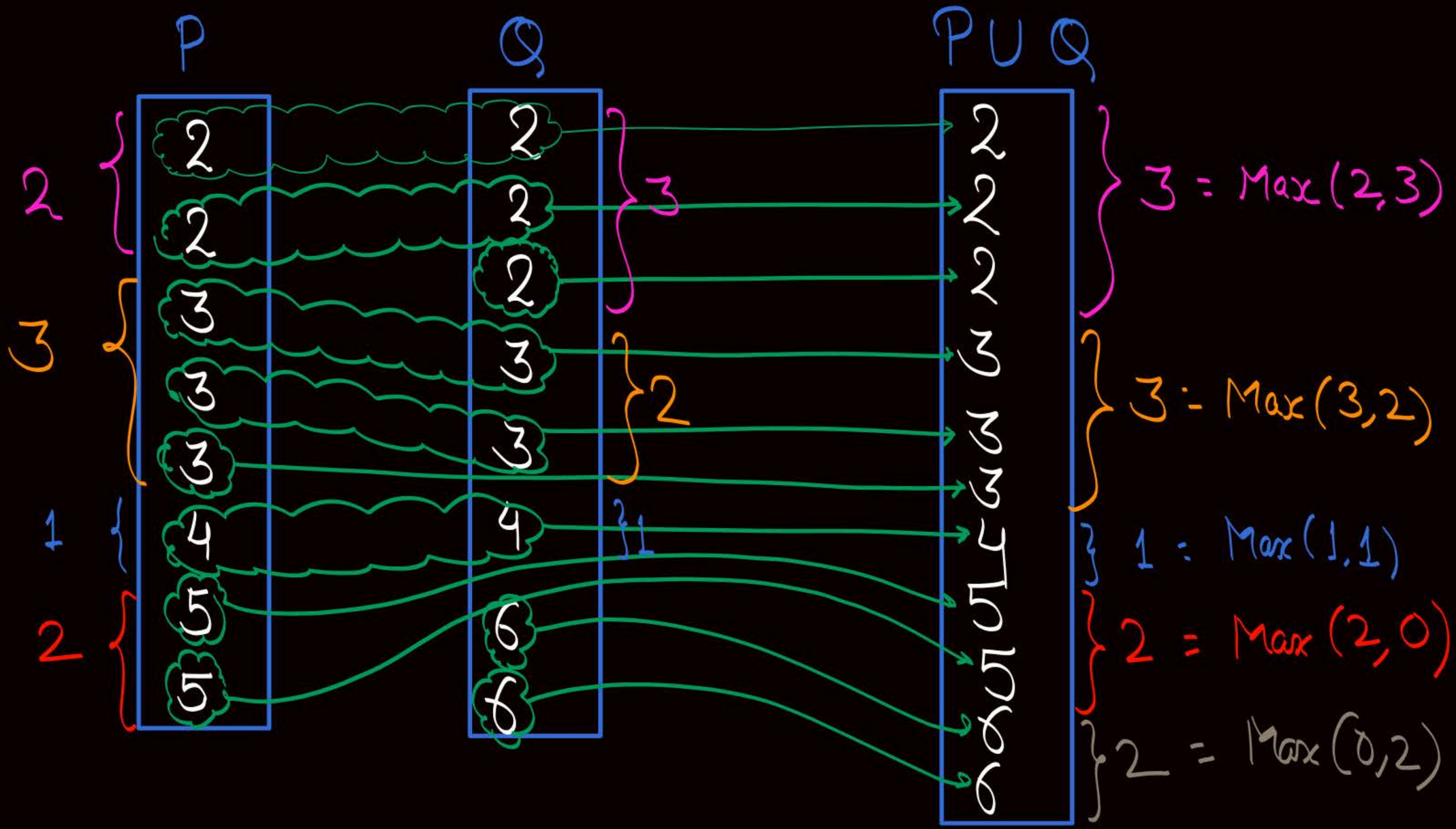
w.r.t multi-sets

(i) $P \cup Q$

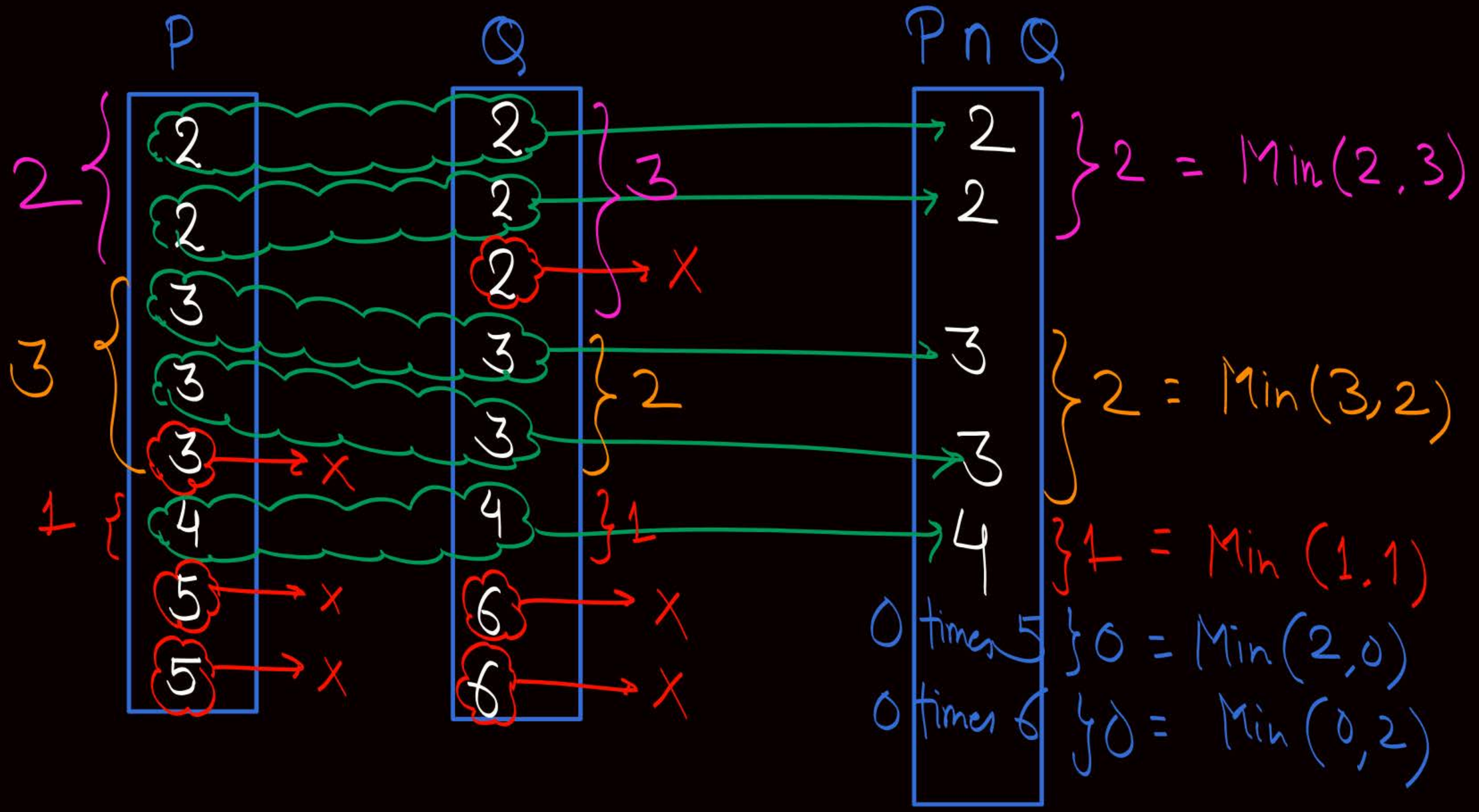
(ii) $P \cap Q$

(iii) $P - Q$

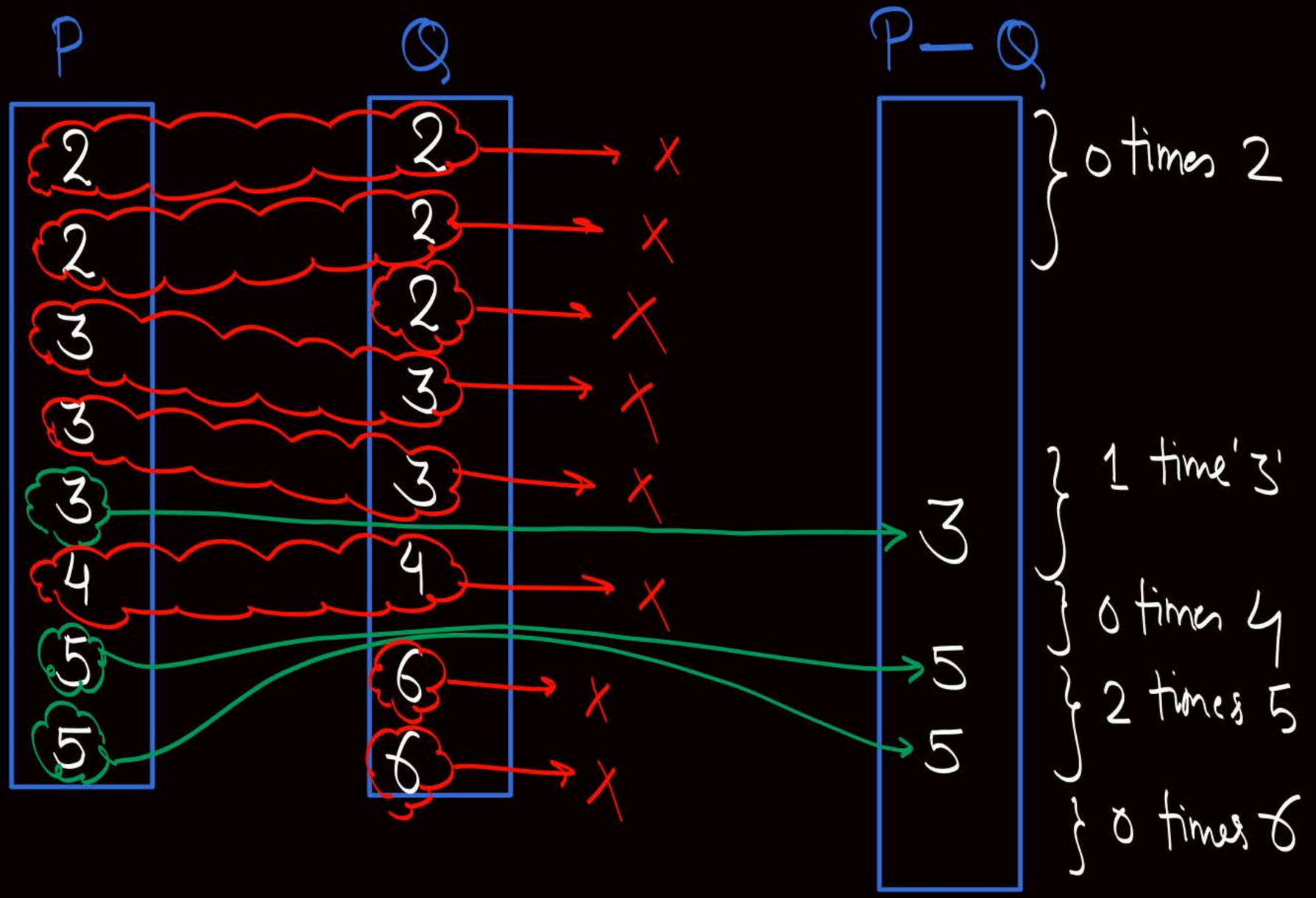
Multiset P ∪ Multiset Q



Multiset P ∩ Multiset Q



Multiset P — Multiset Q



Let multiset $P = \{ m_1.a_1, m_2.a_2, m_3 a_3, \ldots\ldots, m_k.a_k \}$

where 'm_i' is the multiplicity of element a_i.

And

let multiset $Q = \{ n_1.a_1, n_2.a_2, n_3.a_3, \ldots\ldots, n_k.a_k \}$

where 'n_i' is the multiplicity of element a_i.

Find multiplicity of a_i in

Output multi-set

(i) $P \cup Q$ = Multiplicity of a_i in $P \cup Q$ = $\text{Max}(m_i, n_i)$

(ii) $P \cap Q$ = Multiplicity of a_i in $P \cap Q$ = $\text{Min}(m_i, n_i)$

(iii) $P - Q$ = Multiplicity of a_i in $P - Q$ = $\begin{cases} m_i - n_i, & \text{if } m_i > n_i \\ 0, & \text{otherwise} \end{cases}$

Q: Let $A = \{1, 2, 3, 4, \ldots, n\}$

How many different multisets can be created using the element of set A.

{ Any element of set A can be chosen any number of times }

Multisets possible with element '1' of set A.

$M_1 = \{1\}$ Size of $M_1 = 1$

$M_2 = \{1, 1\}$ Size of $M_2 = 2$

$M_3 = \{1, 1, 1\}$ Size of $M_3 = 3$

$M_{100} = \{\overbrace{1, 1, 1, \ldots 1}^{100 \text{ times}}\}$ Size of $M_{100} = 100$

$M_\infty = \{1, 1, \ldots$ infinite times $1\}$ Size of M_∞ = not definite

If we do not have any restriction on the size of multi-set, then infinite multi-sets are possible with a single element.

∴ Ans = infinite

H.W

Q: Let $A = \{1, 2, 3, 4, \ldots, n\}$

How many different multisets of size = 4 can be formulated using the elements of set A, such that at least one element of set A appears exactly twice in each such multiset.

Summation of multiplicities = 4

HoNo Q: Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$

How many different multisets of 'size = 4' can be formulated using the elements of set A.

Relations



2 mins Summary

Topic Properties of set operations

Topic Multi-set

Topic Cartesian product

Slide

THANK - YOU